## TURNING 700 ON:

1. Push Switch (located on back)
   Push → ON
   Push → OFF
2. Depress PRIME, Set RUN Mode.

## MODES OF OPERATION

RUN | LEARN | LEARN & PRINT | LIST PROG.



RUN: For normal operation & program execution.

LEARN: For entering a program into core from the key-board.

LEARN / PRINT: For entering a pro-gram into core from the key-board while simultaneously printing each step on typewriter.

LIST PROGRAM: For listing current program in core on typewriter. Depress GO to list step number and program code.

## 700 PROGRAM CODES

| CODE | KEY |
|---|---|
| 0400 | + DIRECT |
| 0401 | – DIRECT |
| 0402 | × DIRECT |
| 0403 | ÷ DIRECT |
| 0404 | STORE DIRECT |
| 0405 | RECALL DIRECT |
| 0406 | ↔ DIRECT |
| 0407 | SEARCH |
| 0408 | MARK |
| 0409 | GROUP 1 |
| 0410 | GROUP 2 |
| 0411 | WRITE |
| 0412 | WRITE ALPHA |
| 0413 | END ALPHA |
| 0414 | STORE Y |
| 0415 | RECALL Y |

| CODE | KEY |
|---|---|
| 0500 | + INDIR |
| 0501 | – INDIR |
| 0502 | × INDIR |
| 0503 | ÷ INDIR |
| 0504 | STORE INDIR |
| 0505 | RECALL INDIR |
| 0506 | ↔ INDIR |
| 0507 | SKIP IF Y ≥ X |
| 0508 | SKIP IF Y < X |
| 0509 | SKIP IF Y = X |
| 0510 | SKIP IF ERROR |
| 0511 | RETURN |
| 0512 | END PROG |
| 0513 | LOAD PROG |
| 0514 | GO |
| 0515 | STOP |

| CODE | KEY |
|---|---|
| 0600 | + |
| 0601 | – |
| 0602 | × |
| 0603 | ÷ |
| 0604 | ↑ |
| 0605 | ↓ |
| 0606 | ↕ |
| 0607 | \|X\| |
| 0608 | INTEGER X |
| 0609 | π |
| 0610 | $Log_{10}X$ |
| 0611 | $Log_{e}X$ |
| 0612 | $\sqrt{X}$ |
| 0613 | $10^{X}$ |
| 0614 | $e^{X}$ |
| 0615 | 1/X |

| CODE | KEY |
|---|---|
| 0700 | 0 |
| 0701 | 1 |
| 0702 | 2 |
| 0703 | 3 |
| 0704 | 4 |
| 0705 | 5 |
| 0706 | 6 |
| 0707 | 7 |
| 0708 | 8 |
| 0709 | 9 |
| 0710 | SET EXP |
| 0711 | CHANGE SIGN |
| 0712 | DECIMAL POINT |
| 0713 | $X^2$ |
| 0714 | RECALL RESIDUE |
| 0715 | CLEAR X |

## TAPE DRIVE OPERATION

RELEASE | FORWARD | TAPE READY | REWIND



RELEASE: Releases tape cassette for removal.

FORWARD: Moves tape forward when depressed.

TAPE READY: Places head of tape reader in contact with tape.

REWIND: Rewinds tape when depressed.

**LOADING**
**RECORDING**
**PROGRAMMING**
**NON-PROG.**
**ERROR LIGHTS**

### TO RECORD PROGRAM
(Core to Tape)

1. Insert tape cassette. REWIND, (if necessary).
2. Depress TAPE READY.
3. SET PC to first step of program (or PRIME).
4. Depress RECORD PROG.

### TO "LEARN" PROGRAM
(Keyboard Entry)

1. Depress LEARN mode.
2. SET PC to desired step (OR PRIME).
3. Depress keys in program.
4. Depress END PROG as final step.

### TO LOAD PROGRAM
(Tape to Core)

1. Depress RUN mode.
2. Insert tape cassette. REWIND (if necessary).
3. Depress TAPE READY.
4. SET PC to desired step number (or PRIME).
5. Depress LOAD PROG.

### TO RECORD DATA ON TAPE
(Core to Tape)

1. Insert tape cassette REWIND (if necessary).
2. Depress TAPE READY.
3. SET P.C. To step which starts desired register. (See table of core in manual.) END PROG necessary at last step in register to be recorded.
4. Depress RECORD PROG.

## PROGRAMMING KEYS

| | | |
|---|---|---|
| **LOAD PROG** Transfer Program from tape to core. 0513 | **SKIP IF ERROR** Skips 2 steps if condition met 0510 | **MARK** Mark followed by any code sets the flag at that step. 0408 |
| **END PROG** Defines Prog Block. 0512 | **SKIP IF Y ≥ X** Skips 2 steps if condition met 0507 | **RETURN** Branches back from a sub-routine. 0511 |
| **STOP** Stops program Execution. 0515 | **SKIP IF Y = X** Skips 2 steps if condition met 0509 | **GROUP 1** Addresses Peripheral equipment. 0409 |
| **GO** Continues execution of a program after a stop. May be used as "no operation" step. 0514 | **SKIP IF Y < X** Skips 2 steps if condition met 0508 | **GROUP 2** Addresses Peripheral equipment. 0410 |
| | **SEARCH** Search followed by any code branches unconditionally to corresponding mark. 0407 | |

## NON-PROGRAMMABLE KEYS

Clear X & Y sets program counter to 000 and turns error light off.

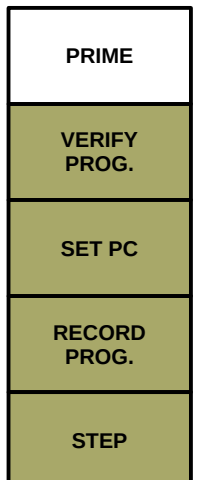
Sums all program codes from step 000 to step preceding END PROG.

Sets program counter with next 3 keystrokes.

Transfers program from core to tape.

In RUN-mode interrupts program execution and allow single step execution of program.

In LEARN mode displays step number and code of each command.




## ERROR LIGHTS
(Sign in display flashing)



### PROGRAM ERROR

Turned on when illegal programming operation is performed or calculation results exceeds $10^{99}$. Can be tested using SKIP IF ERROR decision.



### MACHINE ERROR

Turned on when program has not been transferred properly from tape cassette. REPEAT TAPE OPERATION.

| | |
|---|---|
| **WRITE ALPHA** Sets unit for alpha printout 0412 | **END ALPHA** Ends alpha printout 0413 |
| **WRITE** Used for numeric printout & format control 0411 | $\frac{1}{X}$ 0615 |
| **INTEGER X** Truncates X leaving integer part in X 0608 | **\|X\|** 0607 |
| $10^X$ 0613 | $LOG_{10}X$ 0610 |
| $e^X$ 0614 | $LOG_eX$ 0611 |

| | |
|---|---|
| **RECALL RESIDUE** 0714 | |
| **π** Displays π in X 0609 | |
| ⇄ Interchanges Y with X 0606 | |

## RECALL RESIDUE

Used for double precision arithmetic to obtain another 12 digits of accuracy when adding, subtracting, multiplying. After division, remainder in X.

EXAMPLE: ADD

5 0 2 4 8 7 3 0 5 8 . 2 8

+

6 . 8 5 2 0 9 8 7

+

Read

.502487306513 +10 in Y

RECALL RESIDUE

Read remaining digits

.209870000000 –02 in X

| |
|---|
| ↓ Y into X Y unchanged 0605 |
| ↑ X into Y X unchanged 0604 |

| | | | |
|---|---|---|---|
| **CHANGE SIGN** 0711 | $\sqrt{X}$ 0612 | $X^2$ 0713 | **CLEAR X** 0715 |
| ÷ Y ÷ X → Y X unchanged 0603 | **7** 0707 | **8** 0708 | **9** 0709 |
| × Y × X → Y X unchanged 0602 | **4** 0704 | **5** 0705 | **6** 0706 |
| – Y – X → Y X unchanged 0601 | **1** 0701 | **2** 0702 | **3** 0703 |
| + Y + X → Y X unchanged 0600 | **0** 0700 | **.** 0712 | **SET EXP** Indicates digits to follow affects exponent of X. 0710 |

## NUMERIC ENTRY & BASIC ARITHMETIC

## OPERATIONS USING X & Y

1. $Log_{10}\ 256 =$

Index 2 5 6

$Log_{10}X$

Read 2.40823996532 in X

2. $e^{5.23} =$

5 . 2 3

$e^X$

Read 186.792803518 in X

3. $71 + 22 + Log_e\ 13 =$

Index 7 1 + 2 2 +

1 3 $Log_eX$ +

Read 95.5649493574 in Y

4. $\sqrt{\frac{55 \times .876 \times 10^{-5}}{63}}$

Index 5 5 ↑ 8 7 6 CHS SET EXP 5

× 6 3 ÷ ↓ $\sqrt{X}$

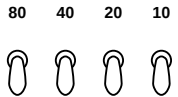
Read .276543288611 –02 in X

Typos/errors in originals, not corrected.

**SPECIAL OPERATION KEYS**

**STORAGE REGISTERS**

## TOGGLE SWITCHES AND SPECIAL OPERATION KEYS




| 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

The Toggle Switches and Special Operation Keys are used for four basic purposes:

1. To address subroutines (such as TRIG and STAT functions) with 1 keystroke.
2. To address 120 data registers.
3. To generate any 700 program code.
4. To format output of typewriter.

---

EXAMPLE: Find SIN 30° (if TRIG PACK loaded)

Index 3 0 **02**

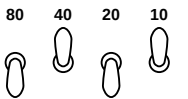
Read .500000000000 in X

NOTE: To protect the loaded functions by pressing VERIFY PROG before learning additional program steps.

122 DATA REGISTERS (000 to 121) Which can be used to add, subtract, multiply, and divide. A register is designated in 1 of 2 ways.

1. Indirect addressing by integer number in Y.
2. Direct Addressing by sum of toggle switches and special operation keys.

**STORE DIRECT**




Depress STORE DIRECT **06**

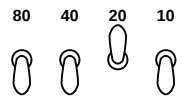
Register 56 is addressed.

| | | | |
|---|---|---|---|
| **RECALL INDIRECT** Recalls to X, REG. indicated in Y 0505 | **INDIRECT** Exchanges X with REG. indicated in Y 0506 | **DIRECT** Exchanges X with REG. designated by next key 0406 | **RECALL DIRECT** Recalls to X, REG. designated by next key 0405 |
| **STORE INDIRECT** Store X into REG. indicated in Y 0504 | ÷ **INDIRECT** Divides X into REG. indicated in Y 0503 | ÷ **DIRECT** Divides X into REG. designated next key 0403 | **STORE DIRECT** Stores X, into REG. designated by next key 0404 |
| | × **INDIRECT** Miltiplies X by REG. indicated in Y 0502 | × **DIRECT** Multiplies X by REG. designated by next key 0402 | |
| | – **INDIRECT** Subtracts X from REG. indicated in Y 0501 | – **DIRECT** Subtracts X from REG. designated by next key 0401 | |
| | + **INDIRECT** Adds X to REG. indicated in Y 0500 | + **DIRECT** Adds X to REG. designated by next key 0400 | |

NOTE: After arithmetic operations (DIRECT & INDIRECT) answer is placed in internal register.

**EXAMPLE:**
Perform 12 × 2
in REG. 21 6

**DIRECT ADDRESSING**
(2 step command)
Toggle Switch Setting




1 2 STORE DIRECT 01
2 × DIRECT 01
6 ÷ DIRECT 01
RECALL DIRECT 01

**INDIRECT ADDRESSING**
(1 step command)
2 1 ↑
1 2 ST INDIR
2 × INDIR
6 ÷ INDIR
RE INDIR

Read 4 in X